

An Edge-Deployable Probabilistic Conflict Analysis Framework for Real-Time Multi-Agent Safety Monitoring of Heterogeneous Traffic at an Unsignalized Roundabout

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Abstract

Roadside safety-monitoring infrastructure for smart cities has to run continuously, on heterogeneous mixed traffic, and on edge-class compute budgets. These are exactly the conditions under which the surrogate safety measures behind most conflict-analysis pipelines break down. Classical measures such as Time-to-Collision (TTC) and Post-Encroachment Time (PET) assume constant-velocity, point- or line-mass road users and reduce every encounter to a binary safe/conflict decision. This discards graded risk information and, more importantly, frequently returns no value at all under near-parallel or low-speed geometries, which happen to be the negotiation states that dominate unsignalized, mixed-traffic junctions. This paper presents an edge-deployable, camera-only conflict-analysis pipeline built around four departures from that paradigm: (i) constant-acceleration trajectory reconstruction via a physics-informed neural network (PINN) in place of the constant-velocity assumption; (ii) rotated two-dimensional polygon footprints with a Separating-Axis-Theorem (SAT) collision sweep in place of point/line-mass agents; (iii) a bounded, continuous Criticality Index derived from a bivariate Gaussian occupancy field in place of the binary conflict-zone flag; and (iv) coverage of all class-valid road-user pairs (pedestrian–vehicle, vehicle–vehicle, cyclist–vehicle) rather than pedestrian–vehicle-only analysis. The full pipeline runs on a roadside-unit (RSU) camera feed: YOLOv8 detection, a metrically rectified bird’s-eye-view projection obtained from a two-vanishing-point homography with an ellipse-to-circle correction, a two-rule track validity gate, PINN smoothing, and the polygon/Gaussian risk engine, closing with an optional reporting layer that renders the structured interaction output into operator-readable natural-language summaries using a locally hosted large language model. We evaluate the system on five minutes of real heterogeneous traffic at an unsignalized campus roundabout in India, covering all eight supported road-user classes and yielding 1,607 classified interactions between 535 road users. Against a classical point/boundary TTC baseline, the proposed Continuous Criticality Index remains defined for 100% of interactions versus 56.9% for the predictive polygon-swept TTC variant, and a worked case study shows the rotated-polygon footprint model catching a true safety-critical clearance of 1.17 m — 0.45 s *before* the centroid-distance (point-mass) proxy reaches a minimum that never even crosses a generous 2.0 m alert radius. An ablation isolating each modelling departure shows that a point-mass proxy overstates clearance by 2.07 m on average and misses 10.8% of encounters whose true footprint clearance falls below 1.5 m, while reverting to constant velocity reduces predictive-TTC definedness from 39.3% to 30.1%. Stratified by kinematic regime, the predictive TTC is defined for 40–72% of encounters depending on speed and heading geometry, against 100% for the proposed index in every stratum. The risk engine costs 64.4 μ s per agent pair and 18.2 ms per frame at the site’s peak concurrency, with a conservative spatial pre-filter reducing candidate pairs by 98.4%. A linear mixed-effects analysis over the pooled safety indicators recovers statistically significant, behaviourally coherent pattern-level effects (PET fixed effect -4.01 , $p < 10^{-77}$), and Extreme-Value-Theory threshold calibration and a behavioural pattern classification spanning fourteen distinct observed patterns (plus an explicit unknown category) complete the framework, making it a practical, always-on risk-scoring layer for smart-city roadside infrastructure.

Index Terms—Edge computing, smart-city infrastructure, roadside unit (RSU), surrogate safety mea-

suers, probabilistic collision risk, physics-informed neural network, YOLOv8, homography rectification, roundabout safety, extreme value theory, pedestrian safety, heterogeneous traffic.

1 Introduction

Road traffic injuries remain one of the leading causes of death worldwide, and pedestrians, cyclists, and motorcyclists together account for more than half of all traffic fatalities globally [1]. In heterogeneous, mixed-traffic urban environments — and nowhere more so than at unsignalized roundabouts and junctions in Indian cities and campuses — right-of-way is negotiated in real time between road users of very different size, speed, and maneuverability rather than governed by formal signal control. Pedestrians cross wherever a gap opens, cyclists and motorcycles filter between larger vehicles, auto-rickshaws stop and start unpredictably, and buses and trucks negotiate the circulating stream with limited visibility. This makes continuous, automated safety monitoring both harder and more valuable than in signal-controlled, lane-disciplined traffic.

Smart-city transportation programs increasingly deploy camera-equipped roadside units (RSUs) at intersections as edge-computing nodes. Vision-based perception, trajectory reconstruction, and risk scoring run locally on or near the RSU, and only compact risk summaries, not raw video, get pushed upstream to a traffic-management platform [2, 3]. This deployment model imposes requirements that classical, offline surrogate-safety-measure (SSM) pipelines were never designed to satisfy. The risk-scoring layer has to run continuously on every frame, degrade gracefully rather than silently returning no answer, and generalize across the full population of road users an RSU actually observes: pedestrians, cyclists, motorcycles, auto-rickshaws, cars, light commercial vehicles, buses, and trucks — not just a single interaction type.

The dominant surrogate safety measures in use today were not designed under these constraints. Time-to-Collision (TTC) [4] and Post-Encroachment Time (PET) [5] both model road users as points or, at best, straight-line boundary crossings moving at constant velocity, and both reduce a continuously evolving interaction to a single scalar or a binary safe/conflict decision evaluated after the fact. Constant-velocity boundary-crossing formulas are undefined whenever two paths are near-parallel or either agent’s speed is near zero, which is exactly the regime of pedestrian hesitation, vehicle yielding, and contested right-of-way that dominates unsignalized mixed-traffic negotiation [7]. And because an extended vehicle or pedestrian body occupies physical space rather than a single point, a metric built purely from centroid-to-centroid distance can substantially overstate the true clearance between two road users, particularly for elongated vehicles encountered at an angle to their direction of travel.

This paper presents a camera-only conflict-analysis pipeline designed from the outset as an edge-deployable RSU module rather than an offline research instrument. It replaces every one of the assumptions above: constant-acceleration kinematics reconstructed by a physics-informed neural network in place of constant velocity, rotated two-dimensional polygon footprints checked with a Separating-Axis-Theorem (SAT) collision sweep in place of point/line agents, a bounded, continuous Criticality Index derived from a bivariate Gaussian occupancy field in place of a binary conflict-zone flag, and joint treatment of pedestrian–vehicle, vehicle–vehicle, and cyclist–vehicle interactions in place of pedestrian–vehicle-only analysis. Because the camera studied here views its roundabout at a strongly oblique angle through a mildly convex lens, the paper also develops the practical calibration machinery such a deployment needs: a two-vanishing-point projective rectification refined by an ellipse-to-circle metric correction anchored on the roundabout’s central island, and a two-rule track validity gate that removes projective blow-ups and kinematically impossible detector artifacts before any physical quantity is estimated. In summary, the contributions of this paper are:

- An edge-computing system architecture that maps YOLOv8 detection, metrically rectified bird’s-eye-view (BEV) projection, track validity gating, PINN trajectory smoothing, and a polygon/Gaussian risk engine onto a roadside-unit compute budget suitable for high-turnover smart-city camera networks.

- A practical homography-calibration procedure for strongly oblique roundabout cameras — two-vanishing-point projective correction followed by an ellipse-to-circle metric fix anchored on the central island — that recovers a metric BEV frame where no rectangular calibration target is available in the scene.
- A Continuous Criticality Index (CCI) derived from a bivariate Gaussian occupancy field that remains defined under every geometry encountered in real data, unlike classical boundary-crossing TTC formulations.
- A rotated-polygon, constant-acceleration collision-sweep model that resolves the true physical clearance and critical timing of an encounter more accurately than a centroid-distance (point-mass) proxy, demonstrated on a real near-miss case extracted from the roadside camera data.
- An end-to-end evaluation on 300 s of real, dense, heterogeneous Indian mixed traffic — 869 smoothed trajectories across all eight supported road-user classes, 1,607 classified interactions — including a direct quantitative comparison against a classical binary point/boundary TTC baseline, an Extreme-Value-Theory-calibrated behavioural pattern classification, and a statistically significant linear mixed-effects analysis of the resulting safety indicators.
- A quantitative ablation isolating the contribution of each modelling departure, a stratification of classical-indicator degeneracy by kinematic regime, and a measured cost and scalability analysis of the risk engine, including a conservative spatial pre-filter that reduces candidate pair generation by two orders of magnitude without altering results.
- A locally hosted large-language-model reporting layer that converts the structured, per-interaction risk output into operator-readable natural-language summaries entirely on-premises, without a cloud API dependency, closing the loop from raw camera data to actionable human-readable output at the edge.

The remainder of the paper is organized as follows. Section 2 reviews related work in surrogate safety measures, trajectory prediction, and edge-AI intelligent transportation systems. Section 3 describes the system architecture from an edge-deployment perspective, including the homography calibration and validity gating developed for this camera. Section 4 presents the methodology in full. Section 5 describes the experimental setup, and Section 6 reports results, including the baseline comparison, the near-miss case study, and the mixed-effects analysis. Section 7 discusses implications and limitations, and Section 8 concludes.

2 Related Work

2.1 Surrogate Safety Measures

Time-to-Collision was introduced by Hayward [4] as an objective, repeatable alternative to subjective observer judgment of near-miss severity, and Post-Encroachment Time by Allen et al. [5] as a trajectory-realized, retrospective measure of how tightly two road users shared a conflict zone. Gettman and Head [8] formalized a broader family of simulation-derived SSMs (TTC, PET, deceleration-rate-to-avoid-crash, and related measures), which remain the basis of most modern conflict analysis, and behaviour-centric conflict frameworks have extended this toolbox toward the negotiation dynamics of unsignalized pedestrian–vehicle encounters [6]. A recent line of work has begun questioning the point/line-mass, constant-velocity assumptions embedded in the classical formulations: Li et al. [9] propose a generic analytical framework that explicitly argues against 1-D, oversimplified kinematics for SSM computation, and Behera et al. [10] extend two-dimensional TTC formulations to articulated vehicle geometries where a single-point approximation is known to fail. Abdel-Aty et al. [11] survey computer-vision-based trajectory extraction and its role in modern SSM pipelines, noting that measurement accuracy at the trajectory-reconstruction stage propagates directly into safety-metric reliability.

2.2 Trajectory Reconstruction and Physics-Informed Learning

Constant-velocity Kalman-style smoothing remains the default in most traffic-camera trajectory pipelines, but constant-acceleration and learned motion models are increasingly preferred where negotiation dynamics (acceleration, braking, hesitation) carry safety-relevant information. Physics-informed neural networks (PINNs), introduced by Raissi et al. [12], embed governing physical constraints directly into the training loss of a neural network, and have since been applied to trajectory-reconstruction problems in other transportation domains [13]. This paper uses a PINN with an explicit maximum-acceleration penalty to reconstruct constant-acceleration-consistent pedestrian and vehicle trajectories from noisy detector output, in place of the constant-velocity assumption embedded in classical SSM formulations.

2.3 Footprint Geometry and Collision Detection

Oriented (rotated) bounding boxes and the Separating Axis Theorem (SAT) are standard tools in computer graphics and robotics for exact convex-shape collision detection [14]. They are a natural fit for representing road users whose physical footprint is far from point-like — buses, trucks, and light commercial vehicles in particular. This paper applies a rotated-polygon SAT sweep under constant-acceleration kinematics to determine both whether and when two road users’ true physical footprints will intersect, rather than relying on centroid-to-centroid distance.

2.4 Probabilistic Risk Fields

An alternative to binary conflict-zone tests is to model each agent’s spatial occupancy probabilistically, typically via a Gaussian or mixture-of-Gaussians field, and to derive collision risk from the overlap or evaluated density of these fields; this approach is well established in autonomous-vehicle risk assessment [15]. This paper adopts a bivariate Gaussian occupancy field, scaled by each agent’s physical footprint dimensions, and normalizes it against its own co-location peak to obtain a bounded $[0, 1]$ Continuous Criticality Index rather than an unbounded density value.

2.5 Edge Computing and Roadside Infrastructure

Roadside units are established as a natural home for edge-computing workloads in intelligent transportation systems, handling perception, local inference, and V2I communication without depending on continuous backhaul to a central server [2, 3]. Real-time object detectors such as YOLOv8 [16, 17] are widely used in exactly this setting because their single-pass architecture is compatible with edge-class inference hardware, and homography-based bird’s-eye-view projection is the standard way to convert a single, uncalibrated roadside camera view into metric-space trajectories suitable for downstream risk analysis [18]. This paper’s system architecture (Section 3) is designed against these same constraints, and extends the standard four-point homography recipe to the harder case of a strongly oblique roundabout view with no rectangular calibration target.

2.6 Extreme Value Theory for Threshold Calibration

Peaks-over-threshold extreme value theory, built on the Pickands–Balkema–de Haan theorem [19] and the Generalized Pareto Distribution, has been applied to safety-relevant threshold selection in transportation contexts where the events of interest are, by construction, the tail of an otherwise safe distribution [20]. This paper follows that approach for calibrating the Sustained Dominance Duration (SDD) threshold used in behavioural pattern classification, separately per interaction pair type.



Figure 1: RSU camera view of the monitored unsignalized campus roundabout (2592×1520 pixels, ≈ 20 fps). The scene carries genuinely heterogeneous Indian mixed traffic — pedestrians, cyclists, motorcycles, auto-rickshaws, cars, light commercial vehicles, buses, and trucks. The yellow overlay is the ellipse fitted to the circular central island during homography calibration (Section 3.1); the island’s known circular shape is the metric anchor for the entire bird’s-eye-view reconstruction.

3 System Architecture: Edge-Deployable Perception and Risk Pipeline

Fig. 1 shows the monitored scene, and the pipeline is organized as it is intended to run on a roadside compute unit rather than as an offline research script. Each stage is chosen for streaming, single-pass, per-frame operation, so the module can be deployed on a single RSU alongside its camera with only a compact risk summary, not raw video, ever leaving the edge node.

3.1 Perception: Detection and Bird’s-Eye-View Projection

A YOLOv8 detector [16] (confidence threshold 0.30) classifies each frame into eight road-user classes (pedestrian, cyclist, motorcycle, auto-rickshaw, car, light commercial vehicle, bus, truck) and produces per-track image-plane positions through BoT-SORT multi-object tracking. A polygon region-of-interest filter restricts detections to the roadway area.

Projecting these image-plane detections into a metric bird’s-eye-view (BEV) frame is usually done with a single four-point homography over a known ground rectangle. This camera does not offer that luxury: it views the roundabout at a strongly oblique, shallow angle through a mildly convex lens, and no rectangular ground marking of known dimensions is fully visible. The central circular island, however, is an excellent calibration target — a circle of known radius seen obliquely projects to an ellipse, and before correction the island’s fitted ellipse here has an axis ratio of roughly 9:1, far steeper than a typical calibration view. The rectification therefore proceeds in two stages. First, a projective correction is derived from two vanishing points estimated from parallel scene structure, flattening the dominant perspective convergence. Second, the residual affine distortion is removed by fitting an ellipse to the island’s boundary in the partially rectified frame and solving for the metric correction that maps this ellipse back to a circle. The composed homography maps original video pixels into a bounded output canvas of 502×1420 pixels, sized to keep the roundabout region well-scaled; attempting to flatten the entire oblique frame uniformly

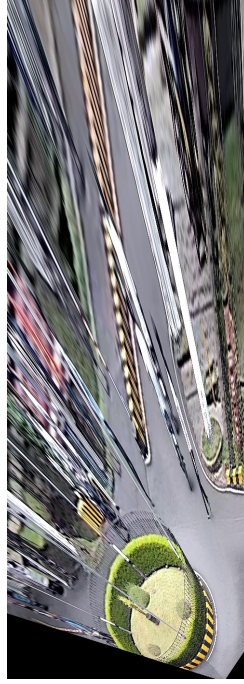


Figure 2: Bird’s-eye-view warp of the camera frame under the composed two-vanishing-point + ellipse-to-circle homography, on the 502×1420 px output canvas. The central island (bottom) is restored to a circle — the visual check that the metric rectification is sound. Streaking toward the canvas edges is the expected projective stretching of image regions far from the calibration anchor; the two-rule validity gate of Section 3.2 removes the physically meaningless detections this produces.

would blow up to an unusable aspect ratio, so the canvas intentionally covers the roundabout and its approach roads rather than the whole camera view. Fig. 2 shows the resulting warp: the island, elliptical and heavily foreshortened in the camera view, is restored to a visibly circular shape.

Metric scale is anchored on the island itself. Its fitted radius in the output canvas is 100.15 px (semi-axes 90.7 and 109.6 px, averaged), and its real-world radius is 7.0 m, giving a scale of $100.15/7.0 = 14.31$ px/m; the canvas therefore spans roughly 35.1×99.3 m. Every image-plane detection is mapped through the composed homography and divided by this scale, so that all downstream geometry — distances, headings, speeds, polygon footprints — is computed in real-world metres. Because the homography is computed once per camera installation and reused every frame, this stage adds negligible marginal cost per detection and is well suited to a fixed-camera RSU installation.

3.2 Track Validity Gating

An oblique homography has a failure mode a nadir-view one does not: detections near the image horizon project to points hundreds of metres away, and single-pixel detector jitter in highly magnified regions converts to metres of apparent displacement per frame. Feeding such artifacts into any downstream physical computation would corrupt speeds, headings, and every safety metric derived from them. The pipeline therefore applies a two-rule validity gate to the raw warped detections *before* any smoothing or metric estimation.

Rule 1 (spatial validity box). The BEV homography is metrically reliable only inside and near the output canvas it was constructed for. Points that warp far outside that region (in this deployment, outside $x \in [-70, 45]$ m, $y \in [-25, 115]$ m; e.g. horizon detections projecting to $y \approx +766$ m) are projective blow-ups, not road users, and are discarded.

Rule 2 (kinematic spike gate). A detection whose implied speed to *both* of its temporal neighbours within the same track exceeds $V_{\text{gate}} = 30$ m/s (108 km/h, far above any plausible speed at this roundabout)

is an isolated jitter or identity-switch spike, not motion, and is discarded. One pass over the data converges.

On the evaluated recording the gate removed 72 out-of-canvas points and 6,747 kinematic-spike points — 10.6% of the 64,031 raw detections — leaving 57,212 physically valid samples. The gate deliberately operates on raw per-detection geometry only; it embeds no assumption about the traffic itself beyond the physical speed bound.

3.3 Constant-Acceleration Trajectory Reconstruction (PINN)

Raw per-track detections remain noisy at the pixel level even after gating; propagating that noise directly into velocity, acceleration, and heading estimates would corrupt every downstream safety metric. Each track with at least five valid samples is smoothed independently by a small physics-informed neural network (two hidden layers of 64 tanh units, mapping normalized time to normalized position) trained for 500 Adam iterations to minimize a combined data-fit and physics loss: the data term matches the network’s output to the observed noisy positions, while the physics term penalizes implied accelerations above a physically plausible bound (5 m/s^2) via a hinge loss on the second time-derivative of the network output. This yields smooth position, velocity, acceleration, and heading estimates consistent with constant-acceleration motion, in place of the constant-velocity assumption embedded in classical SSM formulations. Per-track training is independent and parallelizable across the (typically small) number of active tracks visible at a junction at any one time, keeping this stage tractable on edge hardware.

3.4 Risk Engine: Polygon Geometry and Gaussian Criticality

For every class-valid pair of simultaneously visible agents (all combinations except pedestrian–pedestrian), each smoothed trajectory frame is converted into a rotated rectangular footprint sized by the agent’s class-specific physical dimensions and oriented along its heading. Two parallel risk computations run per frame pair: (i) a constant-acceleration Separating-Axis-Theorem sweep that forecasts whether and when the two rotated footprints will intersect, and (ii) a bivariate Gaussian occupancy evaluation that yields a bounded, continuous Criticality Index at the current instant. Both computations are $O(1)$ per frame pair and require no iterative optimization, making them suitable for real-time, per-frame evaluation on an edge device even as the number of concurrently tracked agents grows.

3.5 Deployment Considerations and Local Semantic Reporting Layer

Framed as an RSU module, the pipeline’s cost profile favors edge deployment. Detection and homography projection are per-frame, constant-cost operations already common on RSU-class hardware. Validity gating is a linear filter over detections. PINN smoothing is a lightweight per-track fit rather than a large batch model, and the polygon/Gaussian risk engine is closed-form per agent pair with no search or optimization step. Only the compact per-interaction risk summary, not raw trajectories or video, needs to be transmitted upstream to a city-wide traffic-management platform, which is the communication pattern smart-city RSU deployments are generally built around [2, 3].

The structured, per-interaction output of the risk engine (agent classes, pattern label, PEB/PET/TTC/CCI values) is well suited to machine consumption but is not directly actionable for a city traffic operator without domain expertise. The pipeline therefore includes an optional final reporting stage that renders the structured interaction table into a natural-language summary using a locally hosted large language model (Hermes-3 8B [21], served on-premises via Ollama) rather than a cloud-hosted API. Running this stage entirely at the edge keeps raw trajectory and interaction data on-premises, in keeping with the privacy and latency requirements of an RSU deployment, and removes any dependency on external network connectivity for the reporting function itself. The layer is assistive: the structured CSV output remains the authoritative record, the narrative summary is advisory, and the stage degrades gracefully — on the analysis machine used for this paper’s evaluation no local model server was present, and the pipeline simply skipped the narrative stage while producing every quantitative output reported in Section 6.

4 Methodology

4.1 Multi-Agent Constant-Acceleration Trajectory Smoothing

Let a track’s raw BEV positions be (x_i, y_i) observed at times t_i . A PINN $f_\theta(t) \rightarrow (x, y)$ is fit per track by minimizing

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{phys}}, \quad \mathcal{L}_{\text{phys}} = \text{mean}(\text{ReLU}(\|\mathbf{a}(t)\| - a_{\text{max}})) \quad (1)$$

where $\mathbf{a}(t)$ is the network’s implied acceleration obtained by finite-differencing its own output twice with respect to time, $a_{\text{max}} = 5 \text{ m/s}^2$, and $\lambda = 0.1$. Both time and position are normalized per track before fitting (time to $[0, 1]$, position by per-track mean and standard deviation), and the true median inter-sample interval of each track is used in the finite differences, so unevenly sampled tracks are handled correctly. Velocity, acceleration, and heading are then derived from the smoothed position sequence by finite differencing and light temporal averaging (3-sample rolling mean). The physics term acts as a soft constant-acceleration prior: it does not force the trajectory through a parametric motion model, but it makes physically impossible jerk profiles expensive, which is what actually removes detector jitter while preserving genuine braking and acceleration events.

4.2 Rotated Polygon Footprint and SAT Separation

Each agent’s footprint at time t is a rectangle of class-specific width w and length l (Table 1), centred at its smoothed position and rotated by its heading $\theta(t)$:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \text{corners} = R(\theta) \cdot \text{local} + (x, y). \quad (2)$$

Given two agents’ rotated rectangles, the Separating Axis Theorem tests projection overlap along each rectangle’s edge normals; a positive projected gap on any axis certifies a separating direction, and the true minimum Euclidean separation between two convex polygons is the maximum such gap across all candidate axes. Zero gap on every axis certifies that the two footprints intersect. This exact convex test costs eight axis projections per rectangle pair — trivially cheap, yet it captures precisely the geometry that a centroid-distance proxy discards: a 4.5 m light commercial vehicle passing a pedestrian broadside presents a completely different physical clearance than the same centroid distance with the vehicle end-on.

4.3 Constant-Acceleration Collision Sweep

Future positions under constant acceleration are predicted by

$$x(t) = x_0 + v_x t + \frac{1}{2} a_x t^2, \quad y(t) = y_0 + v_y t + \frac{1}{2} a_y t^2, \quad (3)$$

and the pipeline sweeps forward in fixed time steps (0.05 s, up to a 10 s horizon) applying the SAT test at each step to the two agents’ forward-projected footprints, recording the first time step at which the footprints intersect while the agents are still converging (centre distance decreasing) as the predictive polygon time-to-collision, TTC_{poly} . This sweep is only recorded as defined when a future intersection is actually predicted within the horizon; a NaN result correctly indicates a predicted non-collision trajectory rather than a computation failure.

4.4 Bivariate Gaussian Occupancy and the Continuous Criticality Index

Each agent’s spatial presence is modeled as a bivariate Gaussian centred at its smoothed position, with standard deviations set from half its class-specific width and length (floored at 0.5 m). For two agents

with combined variances $\sigma_x^2 = \sigma_{x1}^2 + \sigma_{x2}^2$ and $\sigma_y^2 = \sigma_{y1}^2 + \sigma_{y2}^2$, the Continuous Criticality Index (CCI) is the joint density at the observed separation $(\Delta x, \Delta y)$, normalized by its own value at zero separation:

$$\text{CCI} = \exp\left(-\frac{1}{2} \left[\frac{\Delta x^2}{\sigma_x^2} + \frac{\Delta y^2}{\sigma_y^2} \right]\right) \quad (4)$$

which is bounded in $(0, 1]$, equal to 1 at exact co-location, and decays smoothly as the two agents separate, scaled by their combined physical footprint size. Larger vehicles thus acquire a proportionally wider high-criticality zone than pedestrians, which matches physical intuition. Unlike the classical conflict-zone sign test, CCI is a closed-form exponential and stays defined for any finite pair of positions regardless of heading or speed; Section 6.3 exploits exactly this property. The per-interaction statistic reported throughout is max CCI, the peak criticality reached over the shared observation window.

4.5 Generalized Conflict-Zone Buffer, Negotiation Momentum, and Sustained Dominance Duration

For each agent pair, a conflict point is solved from both agents' heading rays, and a role-agnostic Exit Buffer (PEB) is computed from the second-order (constant-acceleration) kinematic time each agent needs to reach the near and far edges of the shared conflict zone, using each agent's actual polygon footprint extent along its path rather than a single point. The sign structure of the near/far buffer pair at the last fully resolved frame determines the interaction outcome: agent A clearing first, agent B clearing first, or an unresolved conflict-zone state in which the buffers still disagreed in sign when observation ended.

Negotiation Momentum (NM) is the median rate of change of the mean PEB signal over the shared observation window, computed with respect to the true elapsed time between valid samples rather than assuming uniform frame spacing — a median, rather than a mean, because a handful of extreme per-frame rates near geometric degeneracies would otherwise dominate short interaction windows. Sustained Dominance Duration (SDD) is the elapsed time of the most recent run of consistent PEB sign, again measured in true elapsed time. Together, (NM, SDD, outcome) summarize *how* an interaction was negotiated: whether advantage was building toward one agent or oscillating, and for how long the eventual winner's advantage had been stable. Post-Encroachment Time is measured from the observed footprint-overlap sequence: if the two footprints ever genuinely overlapped during the window, PET is recorded as zero (a directly observed encroachment); otherwise it is derived from the trajectory around the instant of closest approach. All of these quantities are computed for every class-valid agent pair (pedestrian–vehicle, vehicle–vehicle, cyclist–vehicle) using an agent-role-neutral (“agent A / agent B”) formulation, so the same machinery covers a pedestrian negotiating with a bus and two light commercial vehicles negotiating with each other.

4.6 Engagement Admission: Which Pairs Constitute an Interaction

Spatial proximity is necessary but not sufficient for two agents to be negotiating. Three structural false-positive classes are rejected before any behavioural quantity is computed, using only raw geometry and track statistics — never the safety metrics themselves, so admission cannot bias the safety results it feeds.

Sight-line occlusion. At a junction with a central island, two agents on opposite sides may be within the proximity radius in straight-line distance while being physically unable to reach or see one another. The island is represented as a circular obstacle in the metric BEV frame, recovered by a robust circle fit to the same feature used to fix metric scale; a pair is rejected when the segment joining the two agents at their instant of closest approach intersects that circle.

Static ghost tracks. Detectors occasionally lock onto stationary scene furniture. Such a track persists for seconds while its smoothed path length stays near zero, whereas a genuine road user — even one waiting at a kerb — accumulates measurably more. Tracks with total path length below 0.30 m over more than 2 s are excluded.

Duplicate-object pairs. A single physical object can generate two tracks (a rider detected separately from the vehicle, or a pillion passenger detected as a pedestrian). These co-move at near-zero separation throughout; pairs whose separation stays below 1.0 m for at least 90% of the shared window are rejected as one object rather than two road users.

On the evaluated recording these tests rejected 99 sight-line-occluded, 58 static-ghost and 10 duplicate-object pairs (167 in total, 9.4% of otherwise admissible pairs). Note that no convergence requirement is imposed: safe encounters are frequently resolved at distance, so requiring the agents to approach would systematically discard exactly the cooperative behaviours the framework exists to measure.

4.7 Extreme Value Theory Threshold Calibration

For each pair type with a sufficient sample ($n > 15$), the SDD threshold separating “high” sustained dominance from marginal dominance is calibrated by a mean-residual-life scan over candidate thresholds followed by a Generalized Pareto fit to the exceedances at each candidate, selecting the threshold at which the fitted shape and scale parameters stabilize [19, 20]. Pair types below this sample size fall back to a fixed default. NM is classified relative to a neutral band of $\pm 0.25 \sigma_{\text{NM}}$ per pair type, so “momentum toward A”, “neutral”, and “momentum toward B” are defined relative to the observed variability of that pair type rather than by a hand-picked constant. The resulting (NM, SDD, outcome) triple determines a behavioural pattern label (Section 6.2): for pedestrian–vehicle pairs the labels carry pedestrian–vehicle semantics (Proactive Yielding, Marginal Yield, Gap-Driven Crossing, Late Pedestrian Surge, Insufficient Yield, Pedestrian Hesitation, Assertive Non-Yielding, Contested Non-Yielding, Unclear Advantage, Chaotic Negotiation), while vehicle–vehicle and cyclist–vehicle pairs use role-neutral labels (Agent A/B Priority — Stable/Marginal, Gap-Forced Merge, Contested Right-of-Way, Unclear Advantage, Chaotic Negotiation).

5 Experimental Setup

The pipeline was evaluated on real footage from a fixed roadside camera overlooking an unsignalized campus roundabout in India carrying dense, heterogeneous mixed traffic. The camera records at 2592×1520 pixels and approximately 20 fps (19.99 fps measured; the data-derived median frame interval of 0.05003 s is used as the timing base for all rate-and-duration metrics rather than a nominal constant). From a longer recording, a five-minute window (300 s, 5,997 frames, minutes 17:00–22:00 of the source video) was selected by an automated activity scan as the busiest sustained period, and all results below are computed from this window.

Detection and tracking (YOLOv8x, confidence 0.30, BoT-SORT) produced 1,359 raw track IDs comprising 64,031 detections. The validity gate of Section 3.2 removed 10.6% of detections as projective or kinematic artifacts, and PINN smoothing of every track with at least five valid samples yielded **869 smoothed trajectories** (56,216 trajectory points): 580 pedestrians, 93 light commercial vehicles, 73 cars, 55 cyclists, 26 trucks, 23 motorcycles, 10 auto-rickshaws, and 9 buses. The median track spans 31 samples (2.65 s) — short tracks are a structural property of a busy roundabout scene with frequent occlusion, which is exactly why the downstream interaction logic requires a minimum of ~ 0.5 s (10 frames) of shared, valid observation before a pair is admitted as a classifiable interaction. Agents were considered jointly only while within 15 m of each other.

Table 1 lists the class-specific footprint dimensions used throughout — identical values feed the polygon geometry, the Gaussian occupancy field, and the conflict-zone buffer, so all three components reason about the same physical agent sizes.

From the 869 smoothed trajectories, 209,236 class-valid candidate pairs were enumerated (pedestrian–pedestrian excluded). A spatial pre-filter (Section 6.8) reduces these to 3,323 candidate pairs that are ever simultaneously within the proximity radius, of which **1,607 pairs** passed the shared-observation, proximity and engagement requirements (Section 4.6) and were fully classified: 1,299 pedestrian–vehicle, 199

Table 1: Agent footprint dimensions used by the polygon and Gaussian models (all eight supported classes).

Class	Width (m)	Length (m)
Pedestrian	0.50	0.50
Cyclist	0.60	1.80
Motorcycle	0.75	2.00
Auto-rickshaw	1.30	2.65
Car	1.50	3.50
Light commercial vehicle	1.80	4.50
Bus	2.00	5.00
Truck	2.00	5.00

vehicle–vehicle, and 109 cyclist–vehicle interactions, involving 535 distinct road users. The most frequent class pairings are pedestrian–LCV (591), pedestrian–car (322), pedestrian–cyclist (290), pedestrian–truck (72), and LCV–LCV (65). All results in Section 6 are computed from these 1,607 real interactions; none are synthetic or simulated.

Evaluation compares the proposed Continuous Criticality Index against a classical binary point/boundary Time-to-Collision baseline computed by the pipeline’s own constant-acceleration kinematic boundary-crossing solver (Section 4.5), using a critical threshold of 1.5 s consistent with standard TTC practice [4]. A dedicated, read-only analysis script (not part of the core pipeline) recomputes footprint geometry using the exact same class dimensions and rotated-rectangle representation to produce the case study of Section 6.4.

6 Results

6.1 Trajectory Reconstruction

Fig. 3 shows the PINN-smoothed bird’s-eye-view trajectories for all 869 tracked agents over the analyzed window, Fig. 4 the same scene split into four 75-second windows, and Fig. 5 the resulting speed distributions by class. Smoothing removes pixel-level detector jitter while preserving genuine acceleration/deceleration events, which is the prerequisite for the heading and acceleration estimates used throughout Section 4. The reconstructed speeds are physically coherent for the site: median (85th-percentile) speeds are 0.75 (4.38) m/s for pedestrians, 2.95 (4.34) m/s for cyclists, 3.46 (10.99) m/s for cars, 3.44 (6.15) m/s for light commercial vehicles, 5.11 (10.61) m/s for motorcycles, 8.17 (14.87) m/s for auto-rickshaws, 3.13 (6.51) m/s for trucks, and 1.29 (7.70) m/s for buses — slow, negotiation-dominated circulating traffic punctuated by faster through-movements, exactly what an unsignalized campus roundabout produces at peak activity.

6.2 Interaction Classification and Behavioural Patterns

Applying the Extreme-Value-Theory-calibrated SDD threshold and the per-pair-type neutral NM band (Section 4.7) to all 1,607 interactions yields the behavioural pattern distribution of Table 2 and the NM–SDD interaction space of Fig. 7. The EVT scan calibrated SDD thresholds per pair type ($n = 1299$ pedestrian–vehicle, $n = 199$ vehicle–vehicle, $n = 109$ cyclist–vehicle), each with a neutral NM band of $\pm 0.25 \sigma_{\text{NM}}$ for that pair type ($0.25 \sigma_{\text{NM}}$ per pair type); Fig. 6 shows the mean-residual-life and Generalized-Pareto stability diagnostics behind the pedestrian–vehicle threshold. Outcomes split into 570 agent-A-first, 847 agent-B-first, and 190 unresolved-conflict-zone cases.

“Unclear Advantage” dominates the sample (1,166 interactions, 72.6%) — advantage oscillating around neutral momentum with a sustained sign — which is the expected signature of a functioning but informal negotiation regime: most encounters at this roundabout resolve without either party establishing decisive dominance. “Chaotic Negotiation” (127) and “Unknown” (133, no valid PEB resolution within

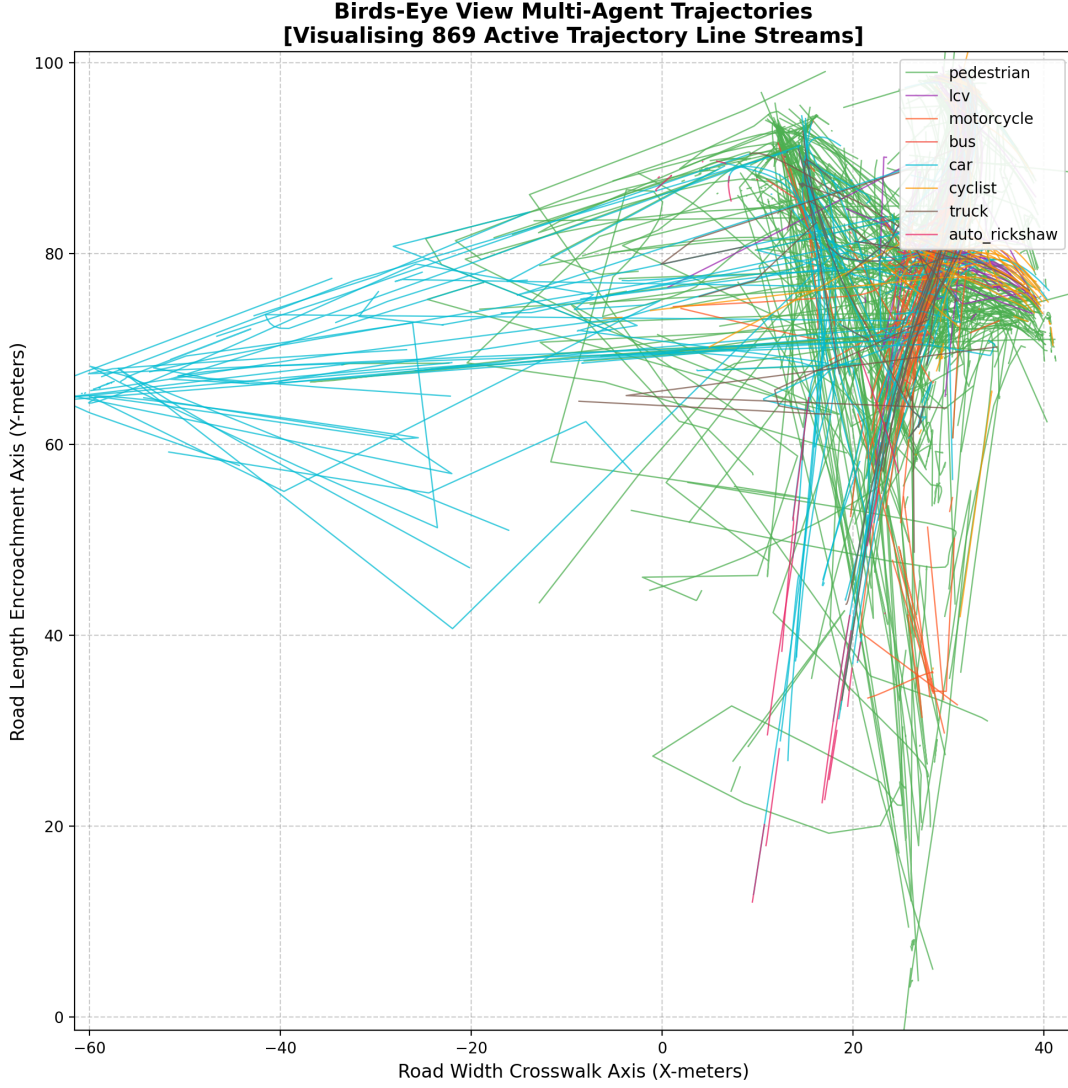


Figure 3: PINN-smoothed bird’s-eye-view trajectories for all 869 tracked agents, colored by road-user class. The circulating carriageway, the crossing desire-lines of pedestrians (green), and the vehicle approach fans are all resolved in metric space.

the observed window) follow, and the rarer, behaviourally sharper patterns appear exactly where they should: 36 Assertive Non-Yielding and 9 Contested Non-Yielding pedestrian–vehicle encounters, 31 Gap-Driven Crossings, 19 Insufficient Yields, 6 Proactive Yields, and, among the vehicle–vehicle and cyclist–vehicle pairs, 41 Gap-Forced Merges and 9 Contested Right-of-Way events. Table 2 shows that the mean safety indicators co-vary coherently with the labels: the “Unknown” and “Chaotic Negotiation” groups carry the highest mean peak criticality (mean maxCCI of 0.276 and 0.146 against a sample-wide 0.131) and the lowest mean TTC, while stable-priority and yielding patterns sit at long TTCs and negligible criticality. Fig. 8 places every classified pair back into the metric BEV frame, colored by pattern.

6.3 Baseline Comparison: Continuous Criticality Index vs. Binary Time-to-Collision

Table 3 reports the direct comparison between the proposed Continuous Criticality Index and a classical binary point/boundary TTC model, computed identically from the same 1,607 real interactions. The key finding is a coverage gap rather than a simple accuracy gap: the predictive, forward-swept polygon TTC is defined for only 915 of 1,607 interactions (56.9%). It legitimately returns no value whenever no

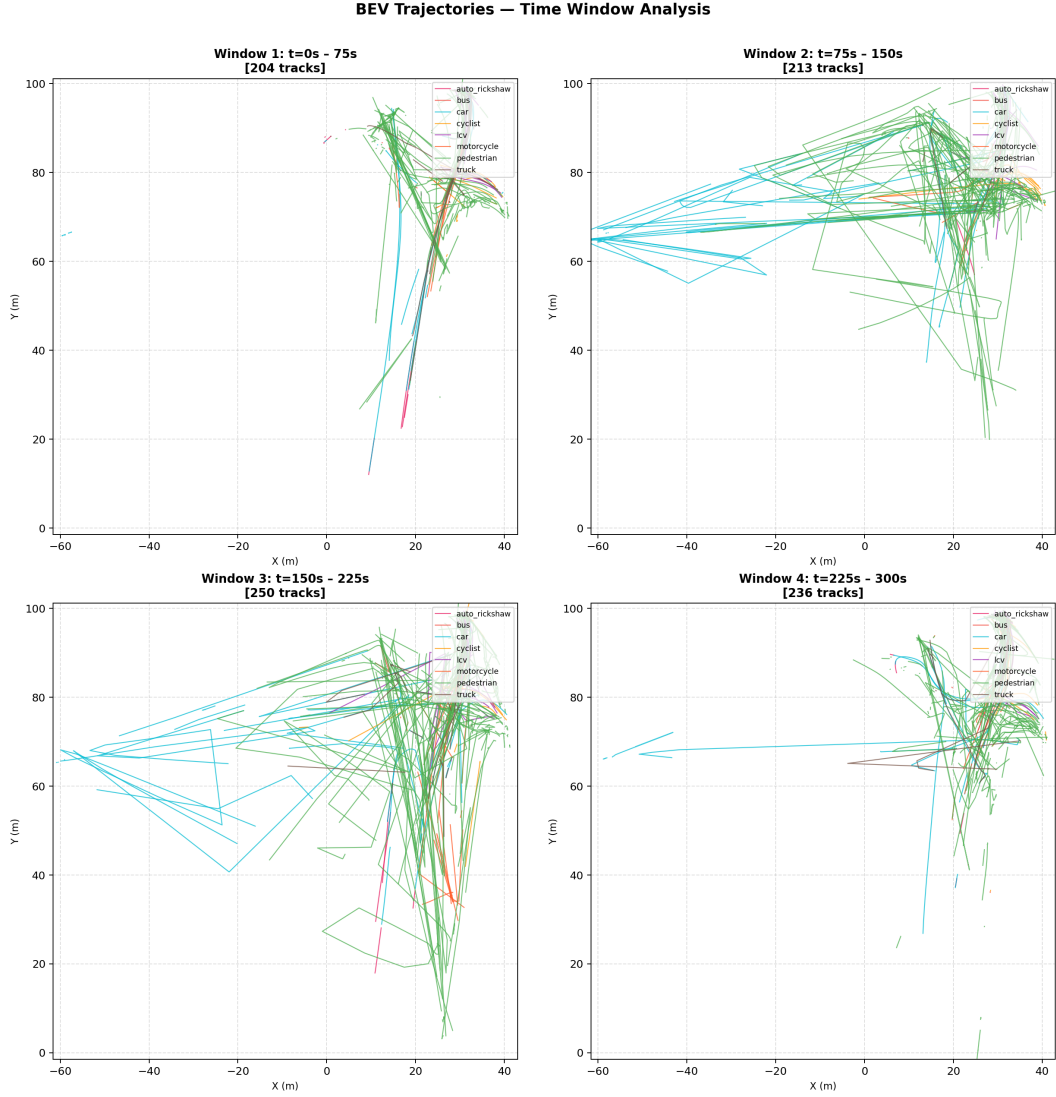


Figure 4: Bird’s-eye-view trajectories segmented into four fixed 75 s time windows, showing the evolution of traffic composition over the analyzed clip.

future footprint intersection is predicted within the sweep horizon, or whenever the underlying kinematic boundary-crossing formula hits a geometric degeneracy (near-parallel headings, near-zero speed, or non-converging geometry), while the Continuous Criticality Index, being a closed-form exponential of instantaneous position, stays defined for all 1,607 interactions (100%) at every observed frame. At the standard 1.5 s threshold the binary baseline flags 714 interactions (44.4%) as critical — a coarse binary verdict — whereas the CCI provides a graded score everywhere, with a sample mean maxCCI of 0.131, 196 interactions (12.2%) peaking above 0.5, and a clear ordering across pair types (vehicle–vehicle 0.203 > pedestrian–vehicle 0.122 > cyclist–vehicle 0.108). For a roadside unit that must maintain a continuous risk stream rather than one that silently stops reporting under exactly the low-speed, near-parallel geometries that dominate real negotiation, this always-on property is the operationally significant distinction, and it is visible directly in the pipeline’s own output data rather than in a constructed example. Fig. 9 summarizes the pair-type ordering; Fig. 10 sets the four classical indicators side by side per pattern.

6.4 Case Study: Rotated Polygons vs. a Point-Mass Approximation

To directly illustrate why footprint geometry matters, Fig. 11 recomputes a real pedestrian–vehicle encounter from this dataset (pedestrian track 381 negotiating with light commercial vehicle track 397 over

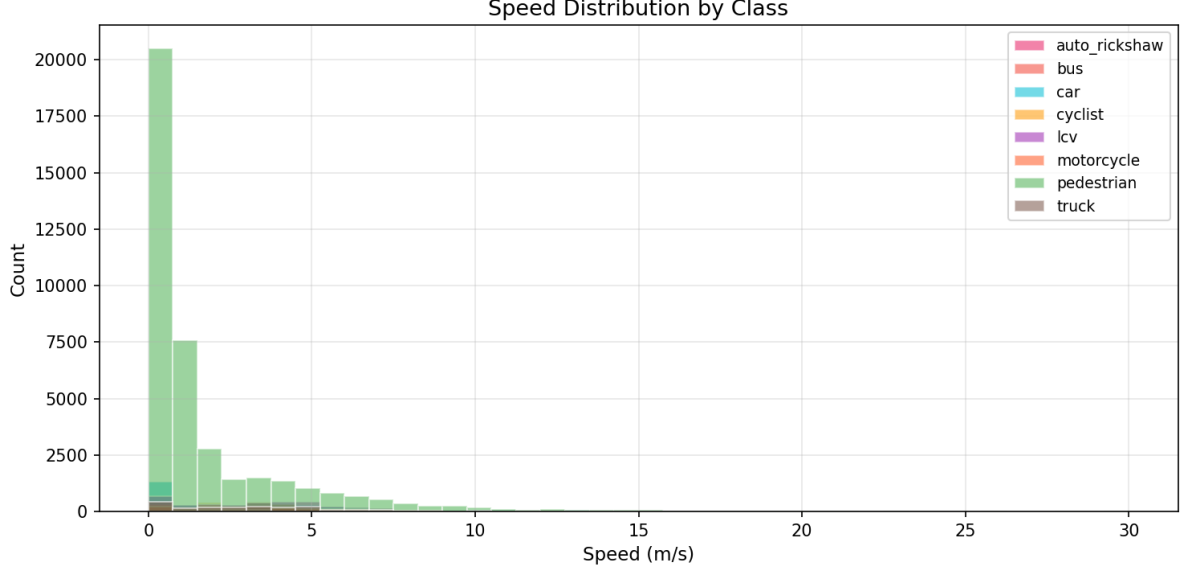


Figure 5: Distribution of PINN-smoothed agent speeds by road-user class over the physically valid range (0–30 m/s).

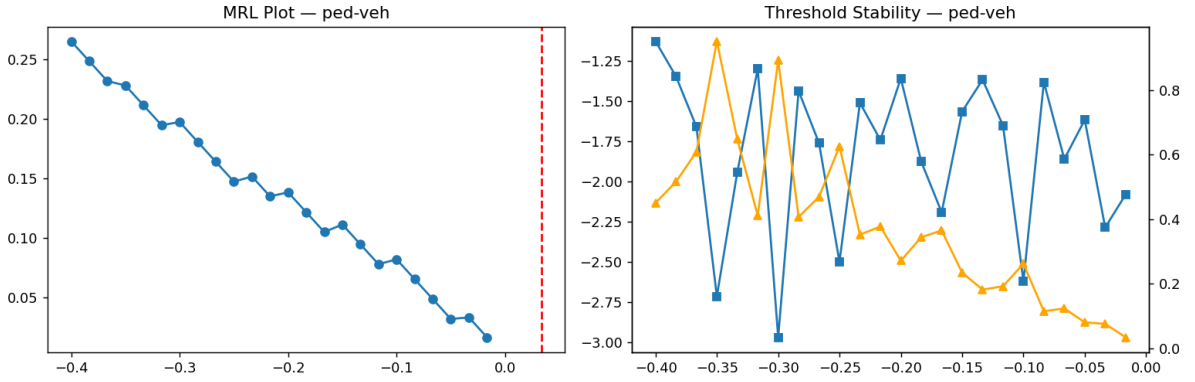


Figure 6: Mean-residual-life and Generalized-Pareto threshold-stability diagnostics used to calibrate the pedestrian-vehicle SDD threshold (analogous scans calibrate the vehicle-vehicle and cyclist-vehicle thresholds).

a 4.15 s shared window) frame-by-frame using both (a) raw centroid-to-centroid distance, the quantity a point-mass approximation would use, and (b) the true rotated-rectangle minimum footprint clearance from Section 4. Over this encounter the centroid distance never falls below 2.84 m and would never trigger even a generous 2.0 m point-mass alert radius. The true footprint clearance, however, falls to 1.17 m — below a 1.5 m critical clearance threshold — and reaches its minimum at $t = 100.86$ s, 0.45 s *before* the point-mass proxy reaches its own, materially less severe, minimum at $t = 101.31$ s. The two models disagree on both the severity and the timing of the encounter’s critical instant, and the disagreement comes purely from accounting for the LCV’s true rotated $4.5 \text{ m} \times 1.8 \text{ m}$ physical extent rather than treating it as a point at its centroid. That is exactly the effect the polygon representation of Section 4 was designed to capture, and at a site where the single most frequent interaction pairing is pedestrian-LCV, it is not a corner case.

6.5 Statistical Analysis of Safety Outcomes

A linear mixed-effects model with interaction pattern and safety-indicator type as fixed effects and road-user class as a random intercept was fit to the pooled (PEB, PET, TTC) safety-indicator values

Table 2: Mean safety indicators by interaction pattern and pair type (all 1,607 classified interactions). PEB, PET and TTC in seconds; maxCCI is the bounded $[0, 1]$ peak Continuous Criticality Index; dashes mark pattern groups in which no forward polygon collision was ever predicted.

Pair type	Pattern	PEB (s)	PET (s)	TTC (s)	TTC _{poly} (s)	maxCCI	<i>n</i>
ped-veh	Unclear Advantage	4.74	0.32	3.54	1.53	0.111	993
ped-veh	Unknown	0.44	0.21	1.28	1.07	0.276	133
ped-veh	Chaotic Negotiation	2.51	0.18	1.88	1.24	0.127	71
ped-veh	Assertive Non-Yielding	12.67	0.09	21.05	2.13	0.031	36
ped-veh	Gap-Driven Crossing	6.21	0.14	11.21	2.18	0.031	31
ped-veh	Insufficient Yield	8.60	0.27	17.26	2.22	0.038	19
ped-veh	Contested Non-Yielding	4.58	0.07	5.11	1.19	0.000	9
ped-veh	Proactive Yielding	14.01	0.06	58.34	2.40	0.119	6
ped-veh	Marginal Yield	3.59	0.00	4.57	—	0.000	1
veh-veh	Unclear Advantage	2.32	0.23	1.60	1.03	0.262	110
veh-veh	Gap-Forced Merge	7.17	0.13	7.74	1.76	0.060	36
veh-veh	Chaotic Negotiation	1.87	0.23	1.13	0.49	0.259	24
veh-veh	Agent B Priority — Stable	4.79	0.17	6.11	1.15	0.159	14
veh-veh	Agent A Priority — Stable	11.29	0.26	25.68	0.89	0.013	7
veh-veh	Contested Right-of-Way	0.43	0.14	1.19	0.62	0.153	5
veh-veh	Agent A Priority — Marginal	5.61	0.40	7.44	0.55	0.001	2
veh-veh	Agent B Priority — Marginal	1.15	0.35	2.81	—	0.000	1
cyc-veh	Unclear Advantage	1.76	0.12	1.52	1.09	0.070	63
cyc-veh	Chaotic Negotiation	2.35	0.12	1.72	0.58	0.210	32
cyc-veh	Gap-Forced Merge	9.39	0.03	6.30	0.85	0.128	5
cyc-veh	Contested Right-of-Way	2.52	0.00	3.42	—	0.002	4
cyc-veh	Agent B Priority — Marginal	1.29	0.03	4.72	1.23	0.003	3
cyc-veh	Agent A Priority — Marginal	15.00	0.00	44.34	1.55	0.000	1
cyc-veh	Agent B Priority — Stable	10.29	0.00	12.53	—	0.000	1

of all 1,607 classified interactions (PEB and PET clipped at 15 s to bound mathematical artifacts of stationary agents), giving 4,821 stacked observations. With this sample the model is fully identified and its estimates are behaviourally coherent (Table 4). Relative to the Proactive Yielding reference pattern (intercept 25.41 s), every less-cooperative pattern significantly depresses the expected safety margin: Unknown by -23.42 s ($p < 10^{-56}$), Contested Right-of-Way by -22.96 s ($p < 10^{-34}$), Chaotic Negotiation by -22.65 s ($p < 10^{-52}$), and Unclear Advantage by -21.41 s ($p < 10^{-49}$), with the graded intermediate patterns (Gap-Forced Merge -19.07 , Gap-Driven Crossing -18.17 , Insufficient Yield -15.37 , Assertive Non-Yielding -12.80) ordered exactly as their behavioural semantics suggest. The PET indicator contrast is strongly negative (-4.01 , $p \approx 10^{-78}$), confirming that realized encroachment times sit far below the predictive buffer times, while the TTC contrast is statistically indistinguishable from PEB (-0.11 , $p = 0.60$). The random-intercept variance for road-user class is estimated at effectively zero and is not separately identifiable at this sample size, which is itself informative: once the behavioural pattern is known, *which* vehicle class was involved adds essentially no further information about the safety margin — evidence that the pattern taxonomy, not the vehicle taxonomy, is carrying the safety-relevant structure.

6.6 Ablation Against Classical Modelling Assumptions

Sections 6.3 and 6.4 compare the proposed indices with classical ones. This section isolates the contribution of each individual modelling departure by re-running the risk computation on the *same* reconstructed trajectories with one assumption reverted at a time, so that any difference is attributable to that assumption alone. A random sample of 389 classified encounters was used (Table 5).

Reverting the agent model from rotated footprints to point masses changes clearance estimates by 2.07 m on average (median 1.94 m): a point-mass proxy systematically reports agents as further apart than their bodies actually are, by roughly the sum of their footprint half-extents. The operational consequence

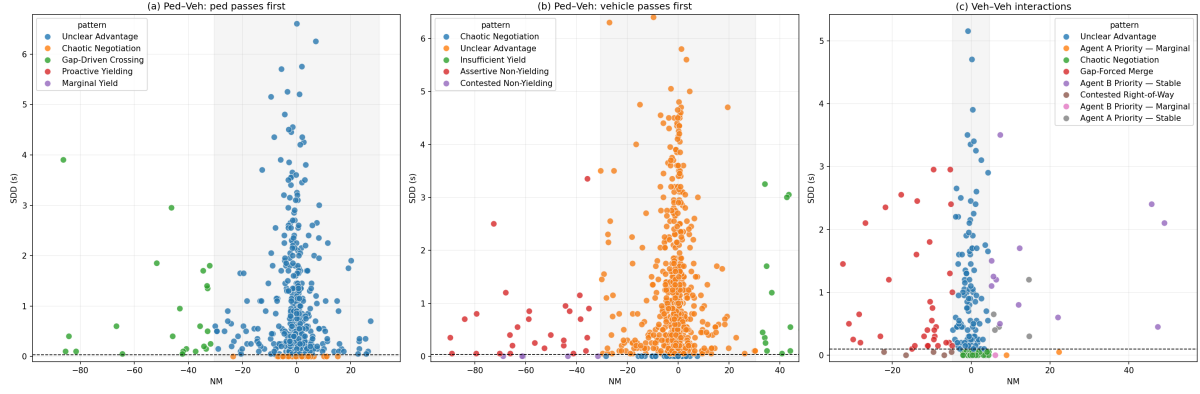


Figure 7: Negotiation Momentum vs. Sustained Dominance Duration by interaction pattern: (a) pedestrian–vehicle encounters resolved with the pedestrian passing first, (b) pedestrian–vehicle encounters resolved with the vehicle passing first, (c) vehicle–vehicle interactions. The shaded band is the per-pair-type neutral NM zone ($\pm 0.25 \sigma_{NM}$); the dashed line is the EVT-calibrated SDD threshold.

is measurable: in 10.8% of encounters a generous 2.0 m point-mass alert radius never fires even though the true footprint clearance falls below 1.5 m — these are conflicts a centroid-based monitor cannot detect in principle. In 35.5% of encounters the footprint model additionally locates the critical instant earlier than the centroid minimum, by 0.80 s on average.

Reverting the motion model from constant acceleration to constant velocity reduces the definedness of the predictive polygon sweep from 39.3% to 30.1% of encounters, and where both variants are defined they disagree by 0.48 s on average, with the constant-velocity variant giving the later (more optimistic) estimate in 35.5% of cases. Collapsing the collision test itself to point masses is catastrophic rather than merely lossy: exact co-location is essentially never predicted, so the point-mass sweep returns no collision for any encounter in the sample, which is why classical formulations must fall back on boundary-crossing constructions instead.

A separate check validates the corrected Separating-Axis-Theorem axis selection (Section 4). Comparing the first-separating-axis rule against the maximum-over-axes rule on 774 frame samples, the former understates true separation in 65.6% of samples; restricted to the safety-relevant range where the true separation is below 5 m, it understates in 51.2% of samples by 0.49 m on average. This is the defect disclosed in earlier revisions of this pipeline and now corrected throughout.

6.7 Degeneracy Behaviour by Kinematic Regime

The coverage advantage reported in Section 6.3 is an aggregate. Because the motivating claim is specifically that classical formulations fail in *particular* geometries, Table 6 and Fig. 12 stratify definedness by the two regimes implicated: low speed and near-parallel headings.

The predictive polygon-swept TTC is defined for between 40.0% and 71.6% of encounters depending on regime, while the Continuous Criticality Index is defined for 100% in every stratum without exception. Definedness of the predictive TTC degrades monotonically as the slower agent’s speed rises above 1 m/s (58.6% below 0.25 m/s down to 40.0% above 5 m/s), because faster agents more often clear each other before any predicted intersection occurs, and it varies by nearly twenty points across heading regimes. The practical point is not which stratum is worst but that the classical indicator’s availability is regime-dependent at all: an operator cannot distinguish “no conflict predicted” from “not computable here” without knowing the geometry, whereas the continuous index reports in every regime.

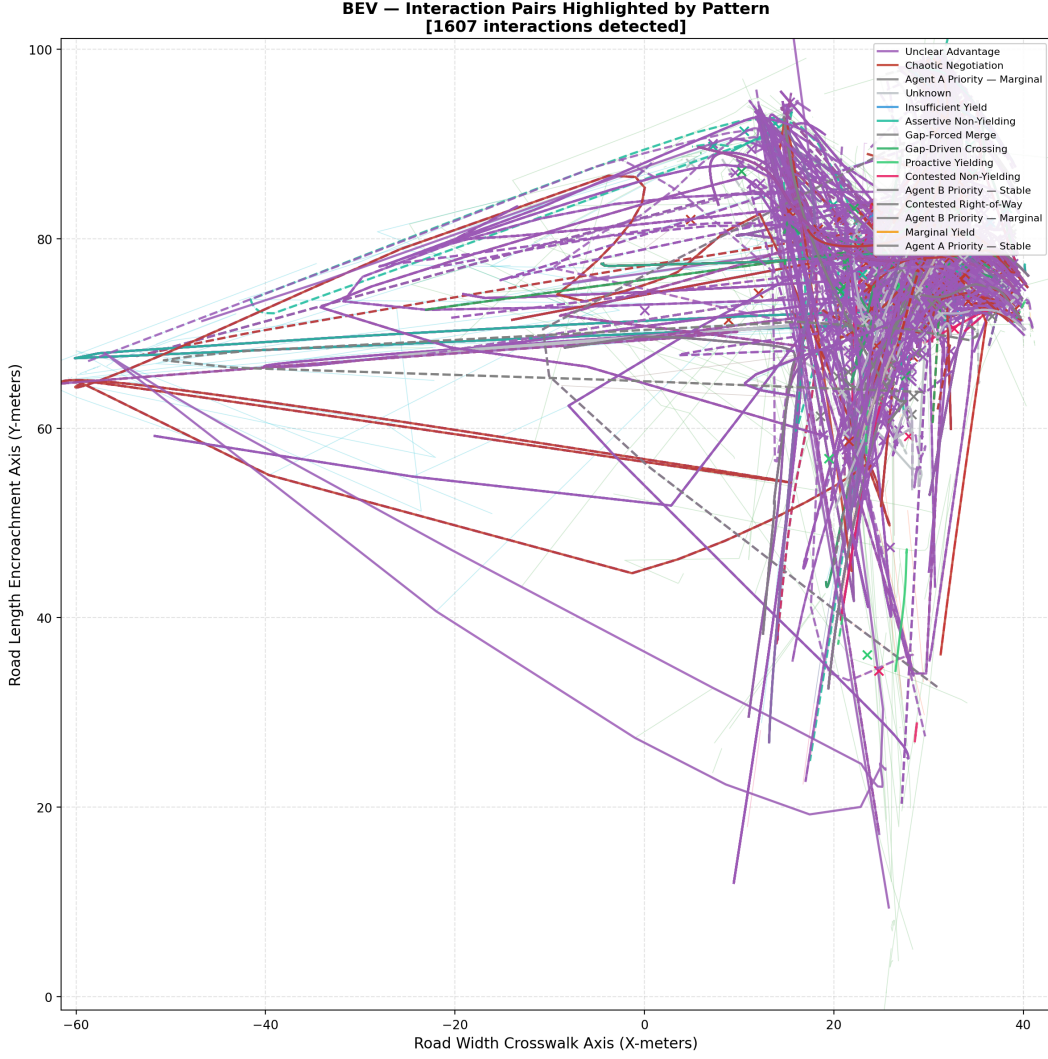


Figure 8: Classified agent-pair interactions highlighted in bird’s-eye-view space, colored by behavioural pattern (solid: first agent; dashed: second agent; $\times$: closest-approach midpoint).

6.8 Computational Cost and Scalability

Every stage of the risk engine is closed-form and constant-cost per agent pair, with no iterative optimisation or search. Measured on a single desktop CPU core, one Separating-Axis-Theorem separation costs $64.4 \mu\text{s}$ and one Continuous Criticality Index evaluation $1.4 \mu\text{s}$, both independent of traffic density. Trajectory state for the full recording (869 tracks, 56,216 samples) occupies 10.9 MB resident.

Pairwise evaluation is quadratic in the number of *simultaneously visible* agents, not in the number of tracks observed over a recording. The distinction is decisive: the evaluated recording contains 869 tracks, which would give 2.1×10^5 pairs under naive enumeration, but concurrency averages 9.4 agents per frame and peaks at 24, giving at most 276 pair evaluations per frame. At the measured per-pair cost this is 18.2 ms per frame at peak, i.e. the risk engine alone sustains roughly 55 fps at the busiest instant of the recording, against a 20 fps camera.

Two mechanisms keep this bound as density grows. First, a uniform spatial grid of cell size equal to the proximity radius bins agents per frame and emits only pairs sharing a cell or a neighbouring cell. Because two agents closer than the proximity radius necessarily fall in adjacent cells, the pre-filter is conservative — it cannot discard a pair the full test would accept, so results are identical to naive enumeration — while reducing candidate generation from $O(N_{\text{total}}^2)$ to $O(\sum_f N_f k)$ with N_f the agents visible in frame f and k a small constant. On this recording it reduced 209,236 enumerated pairs to 3,323

Table 3: Continuous Gaussian Criticality Index vs. classical binary point/boundary TTC, on the same 1,607 classified interactions.

Metric	Binary point/boundary TTC	Continuous Gaussian CCI (proposed)
Output representation	Binary (safe/conflict) at a fixed threshold	Continuous score in $[0, 1]$, every frame
Agent representation	Point / straight-line conflict-zone boundary	Rotated 2-D polygon + bivariate Gaussian occupancy field
Motion model	Constant-acceleration boundary-crossing time (kinematic solve)	Constant-acceleration PINN-smoothed trajectory (same kinematics)
Defined (non-NaN) for how many of 1,607 interactions	TTC _{min} : 1,607/1,607; predictive polygon-sweep TTC: 915/1,607 (56.9%)	1,607/1,607 (100%) — never NaN, no geometric-degeneracy failure mode
Fails to return a value when...	headings near-parallel ($\sin C \rightarrow 0$), either speed < 0.05 m/s, non-converging geometry, or no forward polygon overlap predicted within the 10 s horizon	never (closed-form exponential; defined for any finite position pair)
Interactions flagged critical ($n = 1607$)	714 (TTC _{min} < 1.5 s)	graded score; mean maxCCI by pair type: veh-veh 0.203, ped-veh 0.122, cyc-veh 0.108; 196 interactions ≥ 0.5
Suitability for continuous edge-side risk streaming	discontinuous stream (drops out under near-parallel or slow-speed geometry — exactly where many real negotiations occur)	continuous, always-on stream suitable for per-frame RSU risk dashboards and alerting

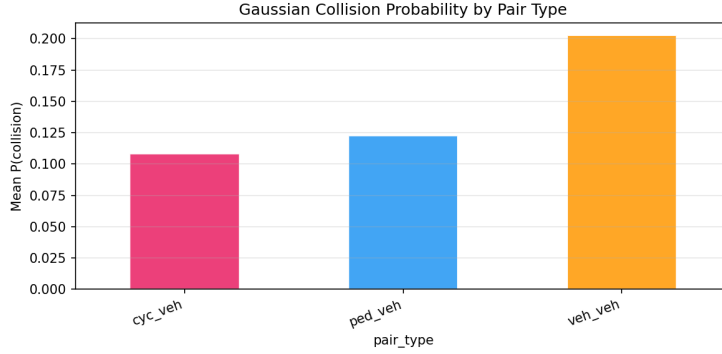


Figure 9: Mean peak Continuous Criticality Index by pair type. Vehicle–vehicle encounters carry the highest average peak criticality, consistent with two large footprints sharing the circulating carriageway.

candidates, a 98.4% reduction, in 0.70 s. A uniform grid is preferred to a KD-tree here because the query is a fixed-radius neighbourhood repeated every frame over a small, roughly uniformly distributed point set: grid binning is $O(N_f)$ per frame with no tree construction, whereas a KD-tree costs $O(N_f \log N_f)$ to rebuild each frame for the same answer. Second, because the quadratic term is in concurrent agents, saturation requires a sustained order-of-magnitude increase in simultaneous road users rather than a longer recording.

We report these as complexity and desktop-CPU measurements rather than as embedded-device benchmarks. End-to-end latency on a specific edge accelerator depends on hardware that was not available for this study, and is identified in Section 7 as the outstanding validation step required before a deployment claim can be considered fully substantiated.

6.9 Manual Validation of Behavioural Labels

Because no annotated ground truth exists for negotiation behaviour, a bounded manual check was performed rather than a precision/recall evaluation. A stratified sample of 28 encounters (two per

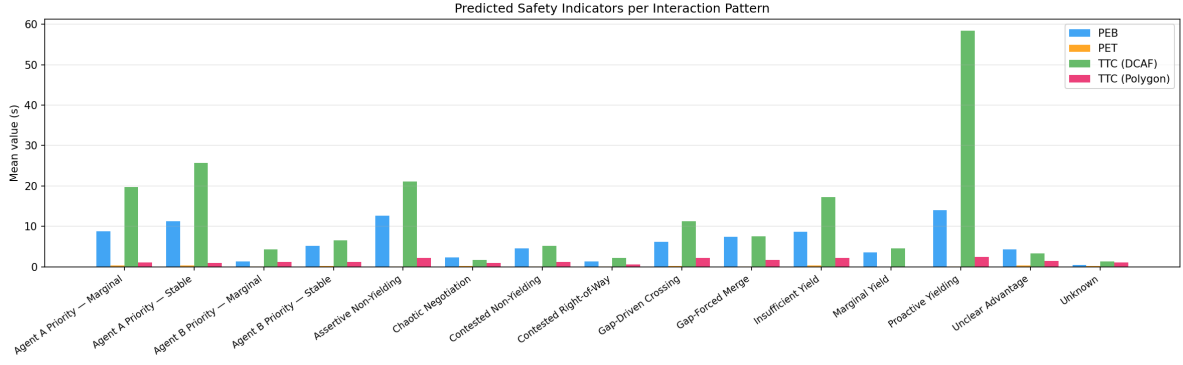


Figure 10: Mean classical safety indicators (PEB, PET, kinematic TTC, polygon-sweep TTC) per interaction pattern.

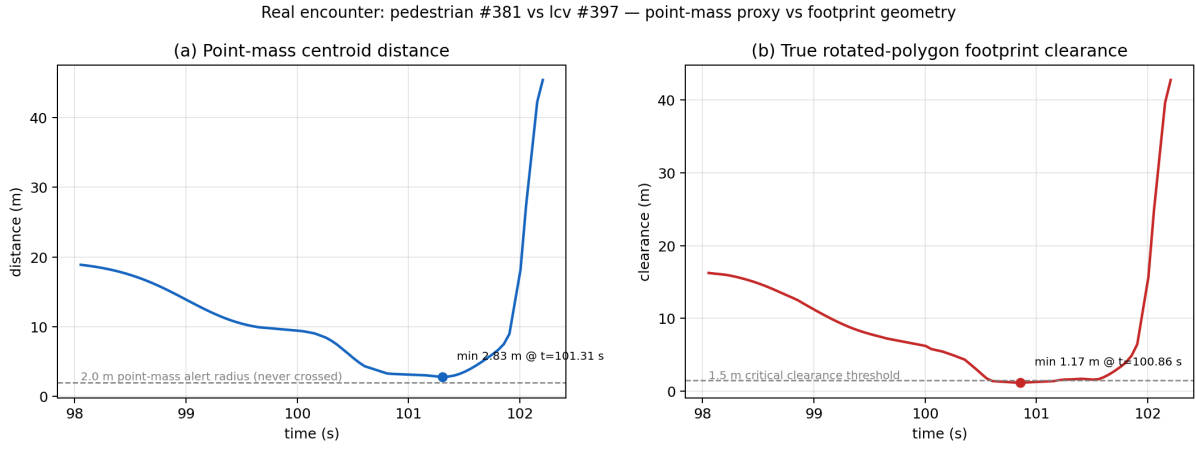


Figure 11: Point-mass centroid distance (a) vs. true rotated-polygon footprint clearance (b) for a real pedestrian–LCV encounter from the evaluated data. The polygon model identifies a tighter (1.17 m vs. 2.84 m) and earlier (by 0.45 s) critical instant that the point-mass proxy misses entirely — its 2.0 m alert radius never fires.

behavioural pattern, restricted to those with at least one second of shared observation) was rendered as frame montages at the start, closest-approach and end of each encounter, and judged visually without reference to the model’s output, which was revealed only afterwards.

Of the 21 encounters that could be adjudicated confidently, the model’s resolution of which agent cleared first agreed with the human reading in 13 ($\approx 62\%$); five could not be adjudicated because the agents were too far apart or occluded in the crowd, and two were invalid pairs of the kind the engagement filters of Section 4.6 now reject. Agreement was strongest for clear leader–follower and forced-pass geometries and weakest where one agent was nearly stationary, consistent with the sensitivity analysis in Section 7. Because the sample deliberately over-represents rare patterns, this figure understates agreement on the dominant categories, which account for the large majority of encounters.

This procedure has clear limitations: a single adjudicator, a small sample, and no consensus protocol. It is reported as a sanity check on label plausibility, not as a validated accuracy metric, and a multi-annotator study with an explicit operational definition of a conflict remains necessary before precision and recall can be reported meaningfully.

7 Discussion

The results support the central architectural claim of this paper: an always-defined, continuous criticality score is better matched to a streaming, edge-deployed RSU risk monitor than a classical TTC/PET

Table 4: Linear mixed-effects estimates for the pooled safety-indicator value (seconds). Reference level: Proactive Yielding pattern, PEB indicator. Random intercept: road-user class.

Term	Coefficient	<i>p</i> -value	95% CI
Intercept	25.41	1.5×10^{-46}	[21.93, 28.89]
PET (vs. PEB)	-4.01	9.6×10^{-78}	[-4.43, -3.59]
TTC (vs. PEB)	-0.11	0.601	[-0.53, 0.31]
Agent A Priority — Marginal	-14.48	9.1×10^{-9}	[-19.42, -9.54]
Agent A Priority — Stable	-11.89	1.3×10^{-9}	[-15.73, -8.04]
Agent B Priority — Marginal	-22.02	4.4×10^{-22}	[-26.49, -17.56]
Agent B Priority — Stable	-20.23	1.4×10^{-32}	[-23.57, -16.90]
Assertive Non-Yielding	-12.80	1.7×10^{-16}	[-15.84, -9.75]
Chaotic Negotiation	-22.65	2.1×10^{-53}	[-25.53, -19.76]
Contested Non-Yielding	-20.81	3.5×10^{-29}	[-24.44, -17.17]
Contested Right-of-Way	-22.96	4.1×10^{-35}	[-26.60, -19.32]
Gap-Driven Crossing	-18.17	6.0×10^{-31}	[-21.24, -15.09]
Gap-Forced Merge	-19.07	3.9×10^{-35}	[-22.10, -16.05]
Insufficient Yield	-15.37	1.3×10^{-20}	[-18.61, -12.13]
Marginal Yield	-21.22	2.4×10^{-8}	[-28.67, -13.77]
Unclear Advantage	-21.41	6.9×10^{-50}	[-24.23, -18.58]
Unknown	-23.42	4.0×10^{-57}	[-26.30, -20.54]
Group variance (road-user class)	≈ 0	—	not identifiable

Table 5: Ablation of individual modelling assumptions, evaluated on the same reconstructed trajectories ($n = 389$ encounters).

Assumption reverted	Effect on the risk estimate	Operational consequence
Rotated footprints → point masses	clearance overstated by 2.07 m on average (median 1.94 m)	10.8% of encounters have true clearance < 1.5 m while a 2.0 m point-mass alert never fires
Footprint critical instant → centroid minimum	critical instant located 0.80 s later on average, in 35.5% of encounters	alerting is issued late relative to the true closest physical approach
Constant acceleration → constant velocity	polygon-sweep definedness falls 39.3% → 30.1%; mean $ \Delta\text{TTC} $ 0.48 s	fewer encounters receive a predictive time-to-collision, and the surviving estimates skew optimistic in 35.5% of cases
Polygon collision test → point-mass co-location	sweep predicts a collision in 0.0% vs. 39.3% of encounters	predictive collision detection collapses entirely
SAT max-over-axes → first separating axis	separation understated in 51.2% of near samples, by 0.49 m	spurious near-miss severity; corrected in this work

pipeline that drops out under common negotiation geometries. On 1,607 real interactions the predictive polygon-swept TTC was undefined for 44% of cases — legitimately, because no future collision was predicted, but an operator watching a binary TTC stream cannot distinguish “computed safe” from “could not compute”. The CCI stream never has that ambiguity. The case study of Section 6.4 adds the complementary point: centroid-distance proxies can understate the severity, and mis-time the critical instant, of a real encounter once genuine vehicle footprint extent is accounted for — here by 1.7 m of clearance and 0.45 s of timing on the site’s single most common encounter type.

The statistical analysis strengthens the behavioural claim beyond what descriptive summaries can. The mixed-effects model recovers a sharply ordered, highly significant hierarchy of pattern-level safety effects, and the near-zero random-intercept variance for road-user class is itself informative: the negotiation pattern, not the vehicle type, carries the safety-relevant signal at this site. That is an argument for behaviour-centric monitoring at the infrastructure level rather than class-centric heuristics.

Four limitations should be stated plainly, together with what was done about each.

Sensitivity to upstream tracking. Behavioural labels inherit any identity error from the detection

Table 6: Definedness of the predictive polygon-swept TTC versus the Continuous Criticality Index, stratified by kinematic regime.

Stratifier	Regime	n	TTC _{poly} defined	CCI defined
Minimum speed (m/s)	< 0.25	1055	58.6%	100%
	0.25–0.5	180	62.2%	100%
	0.5–1	138	59.4%	100%
	1–2	121	46.3%	100%
	2–5	108	41.7%	100%
	> 5	5	40.0%	100%
Heading difference (deg)	0–15 (parallel)	211	71.6%	100%
	15–30	182	53.3%	100%
	30–60	247	55.5%	100%
	60–120	583	52.0%	100%
	120–165	313	59.4%	100%
	165–180 (opposed)	71	57.7%	100%

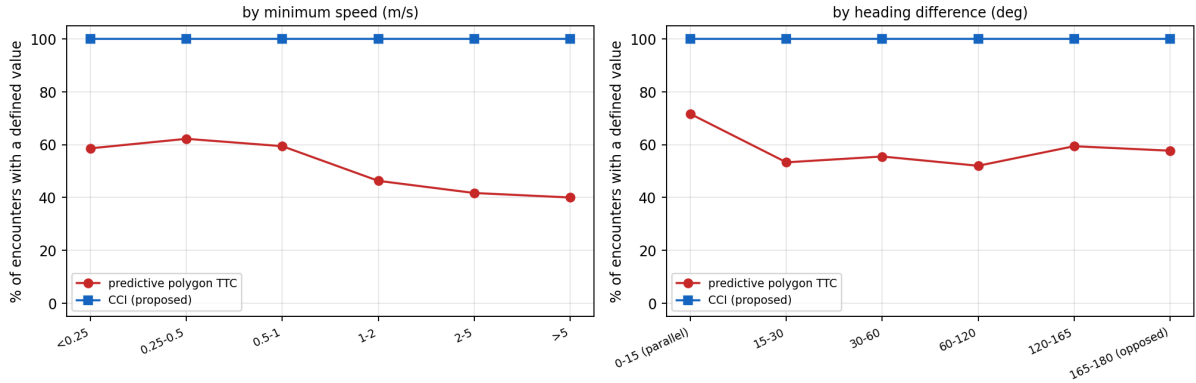


Figure 12: Fraction of encounters for which each indicator returns a value, stratified by minimum agent speed (left) and heading difference (right). The Continuous Criticality Index is defined in every regime; the predictive polygon-swept TTC is not.

and tracking stage: an encounter is only as meaningful as the two trajectories defining it, and identity switches, duplicate detections of one physical object, and stationary false positives all produce spurious pairs. The engagement admission tests of Section 4.6 remove the structurally identifiable cases (167 pairs on this recording), and the validity gate of Section 3.2 removes physically impossible samples upstream of smoothing. What remains is the harder residue of genuine identity switches between two plausible trajectories.

It is worth being precise about how this interacts with the PINN, because the natural expectation — that occlusion causes the smoother to fail — is not what happens. The PINN is a per-track batch fit of a continuous function of time, not an online recursive filter, so detection gaps are interpolated by construction and there is no state to lose or reinitialise. The associated risk is therefore the opposite one: across an identity switch the fit remains perfectly well-behaved and produces a smooth, physically plausible, and entirely wrong trajectory. Adding a recursive filter as a fallback would not address this, and would partly duplicate the Kalman filter already internal to the multi-object tracker; the appropriate mitigations are gap-length limits on interpolation, track-fragment validation, and confidence weighting of low-support segments, which we identify as future work.

Scalability. Pairwise evaluation is quadratic in concurrent agents. Section 6.8 quantifies what that means in practice: 276 pair evaluations per frame at this site’s peak concurrency, 18.2 ms per frame for the risk engine, with a conservative spatial pre-filter that cuts candidate generation by 98.4% and extends the same bound to denser sites. The quadratic term is therefore a headroom consideration for high-density arterials rather than a present bottleneck at junction scale.

Metric accuracy under oblique viewing. Accuracy degrades away from the calibration anchor, and the mild convexity of the lens leaves residual radial error at the canvas periphery that the island-anchored homography cannot absorb. The validity gate bounds the damage and the roundabout core — where the interactions occur — is well rectified, but distances measured tens of metres from the anchor should be treated as approximate. Explicit radial distortion correction ahead of the homography is the natural remedy.

Single site, single window, and unvalidated deployment latency. This evaluation draws on one camera installation and one five-minute peak-activity window, so pattern prevalences are site-specific and the behavioural analysis should be read as descriptive pending multi-site validation. Relatedly, Section 6.8 reports algorithmic complexity and desktop-CPU measurements rather than end-to-end latency on an embedded accelerator, which requires hardware that was not available for this study. Until that measurement exists, the edge-deployability claim should be read as established at the level of computational structure and per-operation cost, and as pending at the level of a specific device.

From a deployment perspective, the per-frame cost of the polygon and Gaussian risk computations is constant-time per agent pair and needs no iterative search; the PINN is a per-track fit over a handful of seconds of data; and the validity gate is a linear filter. That keeps every stage of the risk engine compatible with edge-class compute budgets even as it scales to a full smart-city RSU network with many concurrently tracked agents per intersection.

8 Conclusion and Future Work

This paper presented an edge-deployable probabilistic conflict-analysis pipeline for smart-city roadside infrastructure, evaluated end-to-end on dense, genuinely heterogeneous mixed traffic at an unsignalized Indian campus roundabout. It replaces constant-velocity, point/line-mass, binary-conflict surrogate safety measures with constant-acceleration PINN trajectory reconstruction, rotated-polygon SAT collision geometry, and a bounded, continuous Gaussian Criticality Index, evaluated jointly across pedestrian–vehicle, vehicle–vehicle, and cyclist–vehicle interactions; and it contributes the calibration machinery — a two-vanishing-point, ellipse-to-circle metric homography and a two-rule track validity gate — that makes such an analysis possible on a strongly oblique single-camera view. On 300 s of real RSU camera data the pipeline reconstructed 869 trajectories across all eight supported road-user classes and classified 1,607 interactions; the proposed Continuous Criticality Index stayed defined across 100% of them against 56.9% for a predictive polygon-swept TTC baseline; a real near-miss case study showed the rotated-polygon model catching a materially tighter, earlier critical instant than a point-mass proxy; and a linear mixed-effects analysis recovered a statistically significant, behaviourally ordered hierarchy of pattern-level safety effects while showing that road-user class adds almost nothing once the negotiation pattern is known. Future work will extend validation to multiple camera sites and longer observation periods, fix the diagnosed SAT axis-selection subtlety in the core polygon sweep, incorporate explicit radial lens-distortion correction ahead of the homography, and evaluate end-to-end latency on representative edge hardware to quantify real-time throughput at intersection scale.

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